

# Three Essays on Factor Models

Daniel Abreu

ISEG – Lisbon School of Economics and Management  
Universidade de Lisboa

Supervisors: Prof. João Nicolau & Prof. Paulo Rodrigues

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# Three Essays on Factor Models

This thesis contributes to the literature on factor models by developing new methods for analysing large panels of time series. It introduces approaches that accommodate **nonlinearity** and **nonstationarity**, features widely observed in economic and financial variables.

|     | Essay                                       | Contribution                                                   |
|-----|---------------------------------------------|----------------------------------------------------------------|
| I   | Smooth Transition Factor Model              | Gradual regime shifts in loadings; consistency; linearity test |
| II  | Nonstationary DFM + Threshold Cointegration | Nonlinear VECM in two-level $I(1)$ factor models               |
| III | Financial Cycle Measurement                 | DFM indicator for cyclical systemic risk; CCyB guidance        |

# Essay I

## Smooth Transition Factor Model

# Essay I: Motivation

## Standard assumption

$x_t = \Lambda f_t + e_t$  — factor loadings  $\Lambda$  **fixed** over time

**Problem:** economic relationships shift across business cycle phases

- ▶ Structural break models: one-off, non-recurrent shifts
  - ▶ Threshold factor models: recurrent, but **abrupt** transitions
- ⇒ **This paper:** loadings evolve **smoothly** between two regimes

## This paper: smooth transition factor model

Regime-specific loadings  $\Lambda_1, \Lambda_2$  weighted by a logistic function  
 $g(s_t, \gamma, c) \in (0, 1)$  — **gradual, parametric, recurrent**

## Essay I: Model

### Smooth Transition Factor Model (STFM):

$$\mathbf{x}_t = (\mathbf{I}_N - \mathbf{G}_t)\mathbf{\Lambda}_1\mathbf{f}_t + \mathbf{G}_t\mathbf{\Lambda}_2\mathbf{f}_t + \mathbf{e}_t, \quad t = 1, \dots, T$$

- ▶  $\mathbf{f}_t$ :  $r \times 1$  vector of latent factors;  $\mathbf{\Lambda}_1, \mathbf{\Lambda}_2$ :  $N \times r$  regime-specific loadings
- ▶  $\mathbf{G}_t = g(s_t, \gamma, c) \mathbf{I}_N$ : diagonal transition matrix

$$g(s_t, \gamma, c) = \frac{1}{1 + \exp\{-\gamma(s_t - c)\}}, \quad \gamma > 0$$

- ▶  $s_t$ : observable transition variable;  $c$ : threshold;  $\gamma$ : smoothness

### Interpretation:

- ▶  $g \approx 0$  (when  $s_t \ll c$ ): loadings  $\approx \mathbf{\Lambda}_1$  (regime 1, e.g. expansion)
- ▶  $g \approx 1$  (when  $s_t \gg c$ ): loadings  $\approx \mathbf{\Lambda}_2$  (regime 2, e.g. recession)
- ▶ Nests linear DFM ( $\gamma = 0$ ) and threshold factor model ( $\gamma \rightarrow \infty$ )

# Essay I: Estimation and Theoretical Results

**NLS estimator:**  $\theta = (\gamma, c)$

$$(\hat{\Lambda}, \hat{f}_t, \hat{\theta}) = \arg \min_{\Lambda, f_t, \theta} \frac{1}{NT} \sum_{t=1}^T [\mathbf{x}_t - (\mathbf{I}_N - \mathbf{G}_t) \Lambda_1 \mathbf{f}_t - \mathbf{G}_t \Lambda_2 \mathbf{f}_t]' [\cdots]$$

**Procedure:** profile over a grid  $\Gamma = [0.1, 6] \times [\underline{c}, \bar{c}]$  (trimming  $\tau = 0.15$ ); extract factors via PCA at each grid point; refine with BFGS (5 restarts).

**Theoretical results:**

**Transition parameters**

$\hat{\gamma}, \hat{c}$  consistent as  $N, T \rightarrow \infty$

**Factor space**

PCA estimator consistent at rate  $\min\{\sqrt{N}, \sqrt{T}\}$

Using consistency of  $\hat{\theta}$ , we show the concentrated factor estimator

$\hat{f}_t(\hat{\Lambda}, \hat{\theta}) = N^{-1}[(\mathbf{I}_N - \hat{\mathbf{G}}_t) \hat{\Lambda}_1 + \hat{\mathbf{G}}_t \hat{\Lambda}_2]' \mathbf{x}_t$  recovers the true factor space at the standard rate.

## Essay I: Selection of the Number of Factors

### Key insight (from the structural break literature):

A single structural break in loadings **doubles** the required factor dimension: one set of factors for the pre-break subsample, another for the post-break subsample. The same applies to smooth transitions.

**Consequence:** applying the standard Bai–Ng (2002)  $IC_1$  to a panel with smooth transition dynamics **over-estimates** the number of factors.

**Solution:** we develop a **robustified criterion**  $IC_3$ , adapted to the smooth transition context:

- ▶ Accounts for the effective doubling of the factor dimension under regime shifts
- ▶ Monte Carlo:  $IC_3$  selects  $\bar{r} \approx 2$  (true  $r^0 = 2$ ) across all designs

**Empirical (FRED-QD):**  $IC_3 \Rightarrow \hat{r} = 3$  vs.  $IC_1 \Rightarrow \tilde{r} = 6 \approx 2\hat{r}$ ,

## Essay I: Linearity Test

**Null hypothesis:**  $H_0 : \gamma = 0$  (linear DFM)

**Problem (Davies, 1977):** under  $H_0$ , the threshold  $c$  is **unidentified** — standard LM statistics are non-standard.

**Solution:** Taylor-expand  $g(s_t; \gamma, c)$  around  $\gamma = 0$ :

$$g(s_t; \gamma, c) \approx \frac{1}{2} + \frac{\gamma}{4}(s_t - c) + O(\gamma^3)$$

Under  $H_0$  the nuisance parameter  $c$  **drops out**. The model reduces to testing  $\delta = 0$  in:

$$x_t \approx \Lambda^* f_t + \Delta(f_t \cdot s_t) + e_t$$

The **LM statistic** is asymptotically  $\chi^2$  and can be bootstrapped to account for cross-sectional dependence.

**Empirical result (FRED-QD):**

$LM = 108$     bootstrapped  $p$ -value = 0.006 (1,000 replications)

⇒ **Strongly reject** linearity at all conventional significance levels.



# Essay I: Monte Carlo Results

**Design:** STFM DGP with  $r^0 = 2$ ;  $\gamma \in \{2, 4\}$ ;  $N \in \{25, 50, 100\}$ ;  $T \in \{100, 200, 400\}$ ; 500 replications.

**Transition parameter estimators** ( $\hat{c}$ ,  $\hat{\gamma}$ ): bias close to zero; decreases as  $N, T$  grow — consistent with theory.

**Common component**  $\hat{C}_t = [(I - \hat{G}_t)\hat{\Lambda}_1 + \hat{G}_t\hat{\Lambda}_2]\hat{f}_t$ : RMSE  $\approx 0.42$ – $0.45$  for  $T = 100$ ; stabilises with larger samples.

**Factor number** ( $IC_3$ ): mean  $\approx 2.0$ – $2.4$  across designs (true  $r^0 = 2$ ); stable under both homoskedastic and heteroskedastic errors.

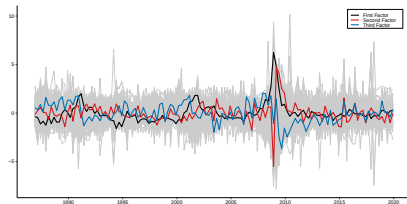
**Linearity test** (rejection rates,  $\gamma = 2$ , homoskedastic):

| $T \backslash N$ | Size (linear DGP) |     |     | Power (STFM DGP) |      |      |
|------------------|-------------------|-----|-----|------------------|------|------|
|                  | 25                | 50  | 100 | 25               | 50   | 100  |
| 100              | 2.6               | 4.6 | 4.4 | 39.6             | 68.2 | 90.4 |
| 200              | 4.0               | 3.8 | 4.2 | 71.6             | 91.8 | 100  |
| 400              | 5.4               | 5.8 | 5.2 | 90.2             | 98.6 | 100  |

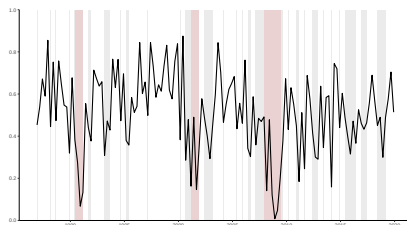
# Essay I: Empirical Application (I) — FRED-QD

**Data:**  $N = 231$  quarterly US series (FRED-QD);  $T = 133$  (1986:Q3–2019:Q4).

$$s_t = \Delta \text{GDP}_{t-1}; \quad \hat{r} = 3 \text{ (IC}_3\text{)}; \quad \hat{c} = -0.005; \quad \hat{\gamma} = 0.955.$$



Black/red/blue: 3 STFM factors. Grey: 231 series.



Transition  $g(s_t, \hat{c}, \hat{\gamma})$ . Light red: NBER recessions.

## Essay I: Empirical Application (II) — Forecasting

### Forecasting equation:

$$y_{i,t+1} = \alpha_i + \sum_{j=0}^q \beta'_{i,j} \hat{f}_{t-j} + \sum_{l=1}^p \delta_{i,l} y_{i,t-l+1} + \varepsilon_{i,t+1}$$

**Out-of-sample relative MSE** vs.  $AR(p)$  benchmark (= 1.000);  
initial window 2012:Q3:

| Model                             | Final Prod.  | Manuf.       | Dur. Cons.   | Dur. Goods   | Cap. Goods   |
|-----------------------------------|--------------|--------------|--------------|--------------|--------------|
| $AR(p)$                           | 1.000        | 1.000        | 1.000        | 1.000        | 1.000        |
| Lin. $r = 1$                      | 0.892        | 1.001        | 0.780        | 0.873        | 0.990        |
| Lin. $r = 2$                      | 0.736        | 0.945        | 0.785        | 0.822        | 0.997        |
| Lin. $r = 3$                      | 0.784        | 1.238        | 1.047        | 0.931        | 1.006        |
| <b>Nonlin. <math>r = 2</math></b> | <b>0.666</b> | <b>0.901</b> | <b>0.734</b> | <b>0.790</b> | <b>0.917</b> |
| <b>Nonlin. <math>r = 3</math></b> | <b>0.663</b> | <b>0.937</b> | <b>0.930</b> | <b>0.880</b> | <b>0.939</b> |

Nonlinear factors (bold) outperform linear counterparts for industrial production and orders.

## Essay II

Nonstationary Dynamic Factor Model  
with Threshold Cointegration

## Essay II: Motivation

This paper contributes to the strand of literature that considers the case in which **factors follow a VECM that admits nonlinear adjustment dynamics**.

- ▶ Empirical evidence suggests that VECM parameters **vary between regimes** (Baumeister & Benati 2013; Liu & Chen 2020)
- ▶ **Threshold cointegration** (Balke & Fomby 1997): adjustments toward long-run equilibrium are activated only when deviations exceed a threshold, capturing frictions, adjustment costs, or market imperfections

### Two specifications:

1. **Threshold factor VECM** — asymmetric dynamics across regimes
2. **Band factor VECM** — adjustment *inactive* within band, *active* outside

Estimated within a **two-level factor model** (global + group-specific I(1) factors), extending Breitung & Eickmeier (2016).

## Essay II: Two-Level Factor Model

**Observation equation for variable  $i$  in group  $r$ :**

$$x_{r,it} = \gamma'_{r,i} \mathbf{G}_t + \lambda'_{r,i} \mathbf{F}_{r,t} + \xi_{r,it}$$

In matrix form:  $\mathbf{x}_{r,t} = \mathbf{\Gamma}_r \mathbf{G}_t + \mathbf{\Lambda}_r \mathbf{F}_{r,t} + \boldsymbol{\xi}_t$

- ▶  $\mathbf{G}_t$ :  $m_0 \times 1$  vector of  $I(1)$  **global factors** (random walk:  
 $\mathbf{G}_t = \mathbf{G}_{t-1} + \mathbf{v}_t$ )
- ▶  $\mathbf{F}_{r,t}$ :  $m_r \times 1$  vector of  $I(1)$  **group-specific factors** (random walk)

**Key question:** Are the stacked group-specific factors  $\mathcal{F}_t = (\mathbf{F}'_{1,t}, \dots, \mathbf{F}'_{R,t})'$  cointegrated, and if so, is adjustment **nonlinear**?

## Essay II: Threshold and Band Factor VECM

Let  $z_{t-1} = \beta' \mathcal{F}_{t-1}$  (cointegration error).

**Threshold factor VECM:**

$$\Delta \mathcal{F}_t = \begin{cases} \alpha_1 \beta' \mathcal{F}_{t-1} + \Phi_1(L) \Delta \mathcal{F}_{t-1} + \mathbf{u}_{1t}, & z_{t-1} \leq \gamma \\ \alpha_2 \beta' \mathcal{F}_{t-1} + \Phi_2(L) \Delta \mathcal{F}_{t-1} + \mathbf{u}_{2t}, & z_{t-1} > \gamma \end{cases}$$

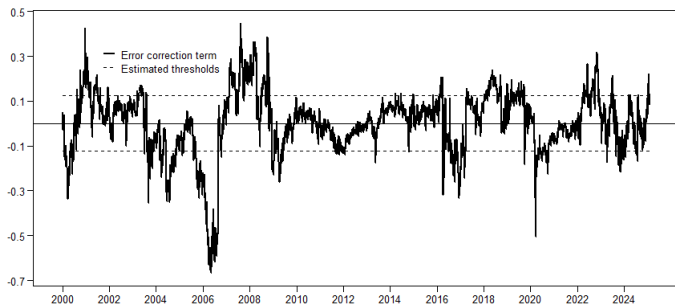
**Band factor VECM** (inactive inside band):

$$\Delta \mathcal{F}_t = \begin{cases} \Phi_2(L) \Delta \mathcal{F}_{t-1} + \mathbf{u}_t, & |z_{t-1}| \leq \gamma \\ \alpha \beta' \mathcal{F}_{t-1} + \Phi_1(L) \Delta \mathcal{F}_{t-1} + \mathbf{u}_t, & |z_{t-1}| > \gamma \end{cases}$$

**Estimation:** SVD initialisation  $\rightarrow$  sequential least squares for factors  $\rightarrow$  joint grid search over  $(\beta, \gamma)$  using reduced-rank regression at each grid point, selecting  $(\hat{\beta}, \hat{\gamma})$  minimising SSR.

## Essay II: Empirical Application — Global Term Structure

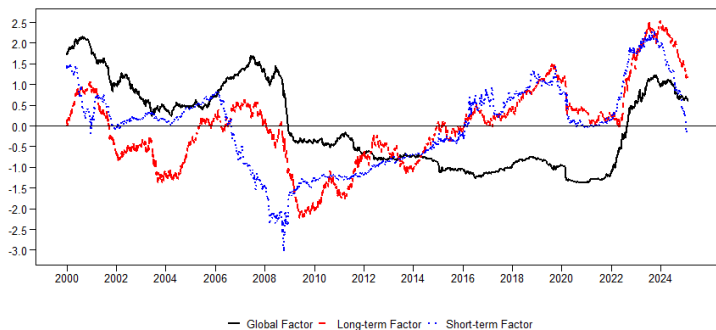
**Data:** 12 advanced economies; 6 maturities (3M, 6M, 1Y, 3Y, 5Y, 10Y)  $\Rightarrow$  72 yields; daily data 2000–2024.



Selected sovereign yield curves. Strong global co-movement motivates a common global factor, with group-specific factors capturing short- vs. long-maturity dynamics.



## Essay II: Empirical Application — Estimated Factors



Top: global factor  $\hat{G}_t$ . Middle: long-maturity group factor  $\hat{F}_{L,t}$ . Bottom: short-maturity group factor  $\hat{F}_{S,t}$ .

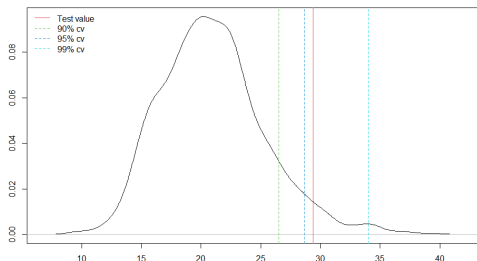
## Essay II: Empirical Application — Testing Linearity

**Step 1 — Johansen trace test** (cointegration between  $\hat{F}_{L,t}$  and  $\hat{F}_{S,t}$ ):

| Test             | Trace stat | 10%   | 5%    | 1%    | Bootstrap $p$ |
|------------------|------------|-------|-------|-------|---------------|
| $H_0$ : rank = 0 | 28.56      | 15.66 | 17.95 | 23.52 | 0.01          |
| $H_0$ : rank = 1 | 5.96       | 6.50  | 8.18  | 11.65 | 0.11          |

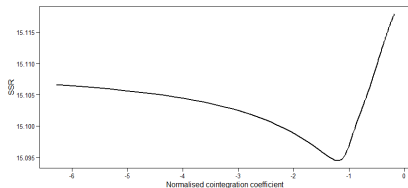
⇒ Reject no cointegration; fail to reject rank 1 at 5%.

**Step 2 — Hansen & Seo (2002) SupLM test:**

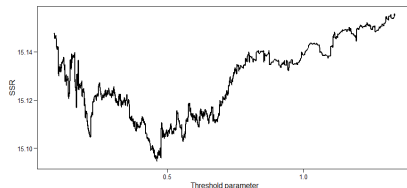


SupLM rejects linear cointegration at 5% ⇒ threshold adjustment present.

## Essay II: Empirical Application — Grid Search



SSR profile over  $\beta$ . Clear minimum at  $\hat{\beta}_{FS} = -1.20$ .



SSR profile over  $\gamma$ . Clear minimum at  $\hat{\gamma} = 0.46$ .

## Essay II: Band Factor VECM Estimation Results

|                                                                | $\hat{F}_{Lt}$      | $\hat{F}_{St}$      |
|----------------------------------------------------------------|---------------------|---------------------|
| Cointegrating vector                                           | 1                   | -1.20               |
| $\hat{\alpha}_1$ (inside band, $ z_{t-1}  \leq \hat{\gamma}$ ) | 0.0044<br>(0.0009)  | 0.0010<br>(0.0009)  |
| $\hat{\alpha}_2$ (outside band, $ z_{t-1}  > \hat{\gamma}$ )   | -0.0006<br>(0.0026) | -0.0061<br>(0.0028) |
| Threshold $\hat{\gamma}$                                       | 0.4622              |                     |

Standard errors in parentheses.

- ▶ **Inside band** ( $\approx 55\%$  of obs): error-correction *inactive* — factors behave as unit roots
- ▶ **Outside band** ( $\approx 45\%$  of obs): error-correction *active*; consistent with **transaction-cost theory**

## Essay III

### Measuring the Financial Cycle: A Dynamic Factor Model Approach

Joint work with Ivan De Lorenzo Buratta (Prometeia)

## Essay III: Motivation

The financial cycle captures “*self-reinforcing interactions between perceptions of value, risk-taking behavior, and financing constraints, which typically manifest as periods of credit booms followed by busts*” (Borio, 2014).

Accurate measurement is key for macroprudential policy, in particular the calibration of the **Countercyclical Capital Buffer (CCyB)**: capital requirements that should *increase* during expansionary phases and be *released* during downturns.

### **Limitations of the Basel III benchmark (credit-to-GDP gap):**

- ▶ Relies on the HP filter ( $\lambda = 400,000$ ) — large end-of-sample bias; upward phase overly incorporated into trend
- ▶ Uses only one variable: misses asset-price and leverage dynamics

**This paper:** A DFM that extracts a single latent factor summarising the co-movement of credit, asset prices, and other financial indicators for 10 European countries.

## Essay III: Data

**10 European countries:** BE, DE, ES, FI, FR, GB, IT, NL, PT, SE

**Sample:** 1995 Q3 – 2024 Q3 (115 observations, quarterly)

### **7 financial indicators:**

| Category             | Variable                                                           |
|----------------------|--------------------------------------------------------------------|
| Credit               | Credit to Non-Financial Corporations (NFC)<br>Credit to Households |
| Asset prices         | House prices<br>Share prices                                       |
| Indebtedness         | Debt Service Ratio (DSR)<br>Credit-to-GDP ratio                    |
| Financial conditions | Government bond spread (vs. Germany, 10Y)                          |

**Transformations:** 2-year year-on-year growth rates (bond spread: 2-year differences); Butterworth filter (exclude periods  $< 4$  years and  $> 25$  years); standardised to mean 0, unit variance.

## Essay III: DFM Framework

### Static factor model (estimated country by country):

$$\mathbf{x}_t = \mathbf{\Lambda} \mathbf{f}_t + \mathbf{e}_t$$

$\mathbf{x}_t = [x_{1t}, \dots, x_{Nt}]'$ :  $N = 7$  standardised financial indicators;  $\mathbf{f}_t$ :  $r \times 1$  latent factors;

$\mathbf{\Lambda}$ :  $N \times r$  factor loadings;  $\mathbf{e}_t$ : idiosyncratic, uncorrelated with  $\mathbf{f}_t$ .

**Factor dynamics:**  $\mathbf{f}_t = \mathbf{\Phi}_1 \mathbf{f}_{t-1} + \dots + \mathbf{\Phi}_p \mathbf{f}_{t-p} + \boldsymbol{\eta}_t$  ( $p = 4$  lags by IC)

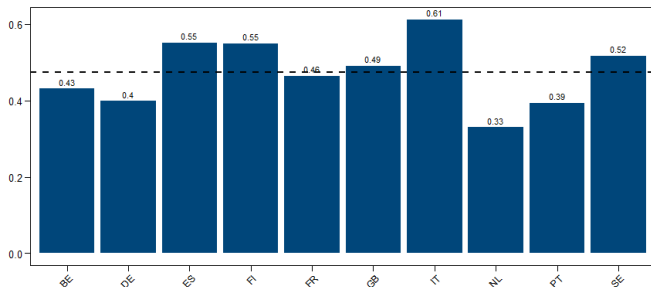
### Three estimation methods:

1. **PCA**: minimises  $\frac{1}{NT} \sum_t (\mathbf{x}_t - \mathbf{\Lambda} \mathbf{f}_t)' (\mathbf{x}_t - \mathbf{\Lambda} \mathbf{f}_t)$
2. **QML**: Kalman filter + EM algorithm (state-space representation)
3. **Two-step (TSTEP)**: PCA initialisation  $\rightarrow$  Kalman smoother

Single factor ( $r = 1$ ) used in baseline; justified by likelihood-ratio test.



## Essay III: Results — Variance Explained



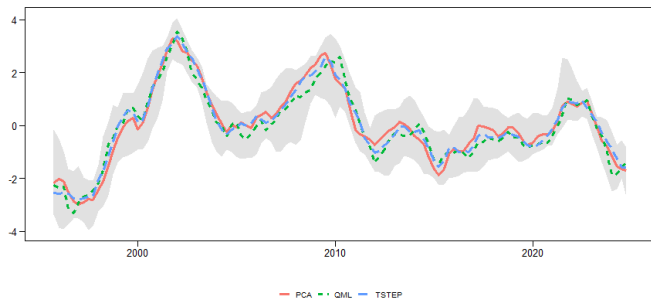
Percentage of total variance explained by the first factor, by country. First factor accounts for approximately **47% of variance on average**, exceeding 50% in several countries.

## Essay III: Results — $R^2$ of the First Factor Across Variables

|            | Credit<br>NFC | Credit<br>HH | House<br>prices | Share<br>prices | DSR         | Credit<br>to-GDP | Bond<br>spread |
|------------|---------------|--------------|-----------------|-----------------|-------------|------------------|----------------|
| BE         | 0.89          | 0.03         | 0.00            | 0.26            | 0.90        | 0.87             | 0.08           |
| DE         | 0.66          | 0.21         | 0.14            | 0.14            | 0.80        | 0.80             | 0.05           |
| ES         | 0.92          | 0.84         | 0.36            | 0.04            | 0.82        | 0.79             | 0.09           |
| FR         | 0.84          | 0.50         | 0.19            | 0.24            | 0.70        | 0.68             | 0.11           |
| IT         | 0.87          | 0.78         | 0.66            | 0.24            | 0.74        | 0.74             | 0.26           |
| PT         | 0.88          | 0.31         | 0.01            | 0.17            | 0.60        | 0.69             | 0.09           |
| ES         | 0.82          | 0.51         | 0.01            | 0.37            | 0.78        | 0.81             | 0.33           |
| <b>Avg</b> | <b>0.78</b>   | <b>0.45</b>  | <b>0.21</b>     | <b>0.17</b>     | <b>0.77</b> | <b>0.76</b>      | <b>0.17</b>    |

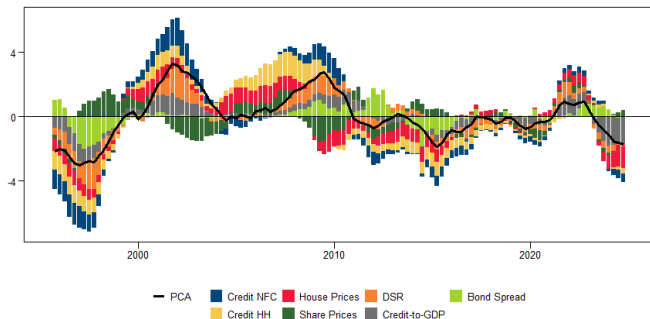
The factor accounts for significant variation in **Credit NFC**, **DSR**, **Credit-to-GDP** — strong co-movement among credit and leverage. More limited explanatory power for share prices and bond spreads.

## Essay III: Results — Financial Cycle Indicators



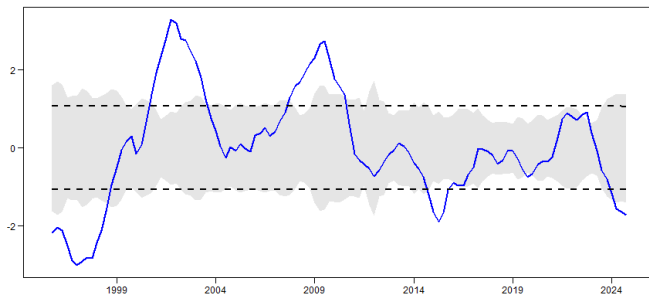
Median of the financial cycle across countries (PCA, QML, TSTEP). Relatively long fluctuations ( $\approx 10$ -year swings); three prominent peaks: early 2000s, GFC (2007–08), post-COVID-19 period.

## Essay III: Results — Variable Decomposition



Decomposition of the median financial cycle across countries. Credit growth and leverage remain primary drivers; asset prices and spreads play significant roles during specific periods.

## Essay III: Neutral Risk Environment



Financial cycle (median) with 80% asymptotic confidence bands derived from distribution theory of the PCA estimator. **Elevated**: above upper band. **Neutral**: within bands. **Subdued**: below lower band. As of 2024 Q3: subdued regime for sample median.

## Essay III: Early-Warning Properties

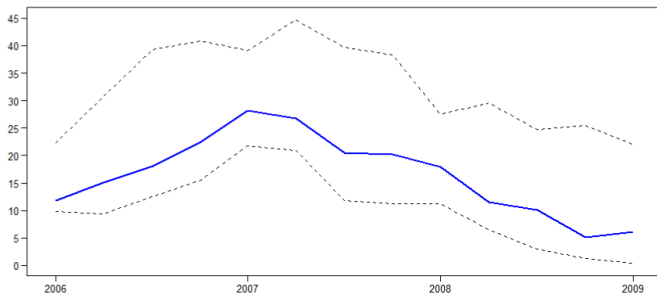
**In-sample exercise:** vulnerability indicator = 1 during 5–12 quarters before systemic banking crisis.

| Country    | AUROC       |             |             |             | Usefulness ( $\theta = 0.5$ ) |             |             |             |
|------------|-------------|-------------|-------------|-------------|-------------------------------|-------------|-------------|-------------|
|            | PCA         | QML         | TSTEP       | BG          | PCA                           | QML         | TSTEP       | BG          |
| BE         | 0.54        | 0.56        | 0.55        | 0.58        | 0.20                          | 0.20        | 0.21        | 0.29        |
| DE         | 0.65        | 0.50        | 0.43        | 0.46        | 0.17                          | 0.14        | 0.06        | 0.18        |
| ES         | 0.94        | 0.85        | 0.91        | 0.51        | 0.45                          | 0.40        | 0.43        | 0.23        |
| IT         | 0.90        | 0.98        | 0.92        | 0.53        | 0.37                          | 0.47        | 0.37        | 0.26        |
| PT         | 0.52        | 0.60        | 0.56        | 0.51        | 0.23                          | 0.25        | 0.26        | 0.26        |
| <b>Avg</b> | <b>0.66</b> | <b>0.63</b> | <b>0.65</b> | <b>0.56</b> | <b>0.25</b>                   | <b>0.24</b> | <b>0.24</b> | <b>0.27</b> |

BG = Basel gap. FI excluded (no crisis in sample). Full table: backup slide.

**Average AUROC: DFM 0.66 vs. Basel gap 0.56.**

## Essay III: Out-of-Sample Early Warning — GFC



Simulated out-of-sample predicted probability of a financial crisis within 1 year. Median crisis probability rises from  $\approx 11\%$  in 2006 Q1 to  $\approx 26\%$  in 2007, providing an early signal ahead of the GFC.

# Conclusions



# Contributions of the Thesis

## Essay I — Smooth Transition Factor Model

Consistent NLS+PCA estimator for

$\mathbf{x}_t = (\mathbf{I}_N - \mathbf{G}_t)\mathbf{\Lambda}_1\mathbf{f}_t + \mathbf{G}_t\mathbf{\Lambda}_2\mathbf{f}_t + \mathbf{e}_t$ ; linearity test;  $IC_3$  factor selection. Nonlinear factors improve FRED-QD forecasts ( $N = 231$ ,  $T = 133$ ).

## Essay II — Nonstationary DFM with Threshold Cointegration

Threshold and band factor VECM in two-level  $I(1)$  models; joint grid search over  $(\beta, \gamma)$ . Global yield-curve factors show discontinuous adjustment ( $\hat{\gamma} = 0.46$ ,  $\approx 55\%$  inside band).

## Essay III — Financial Cycle Measurement

DFM indicator (7 variables, 10 EU countries, 1995–2024); AUROC 0.66 vs. Basel gap 0.56; 80% confidence bands define neutral CCyB environment; out-of-sample crisis probability signals GFC well in advance.

# Common Thread and Future Research

**All three essays relax a restrictive assumption in the standard linear DFM:**

| Essay | Assumption relaxed | How                                      |
|-------|--------------------|------------------------------------------|
| I     | Constant loadings  | Logistic transition matrix $G_t$         |
| II    | Linear adjustment  | Threshold / band VECM on $\mathcal{F}_t$ |
| III   | Single indicator   | Multi-variable latent factor             |

**Future research:**

- ▶ **Asymptotic distribution theory** for the NLS estimators in Essays I and II (confidence intervals for  $(\hat{\gamma}, \hat{c})$  and  $(\hat{\beta}, \hat{\gamma})$ )
- ▶ **Impulse response functions** under nonlinear factor VECMs
- ▶ **Multiple regimes** ( $K > 2$ ) with endogenous regime selection
- ▶ **Real-time financial cycle monitoring** with mixed-frequency data

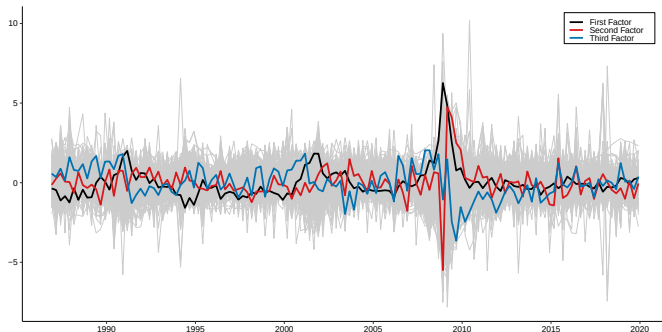
# Thank You

Questions?

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Daniel Abreu  
ISEG – Universidade de Lisboa

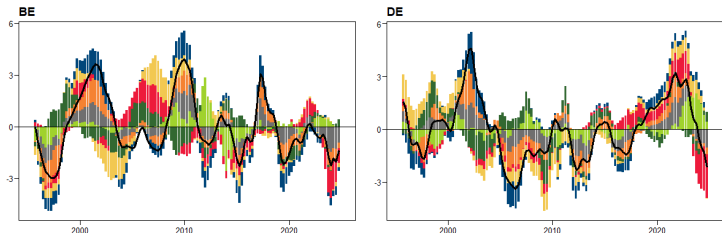
## Backup: STFM — Factor Estimates (FRED-QD)



Black, red, blue: first three STFM factors. Grey background: 231 standardised series.

$N = 231$ ,  $T = 133$  (1986:Q3–2019:Q4);  $\hat{r} = 3$ ;  $IC_3$  criterion.

## Backup: FC — Financial Cycle by Country (1–4)



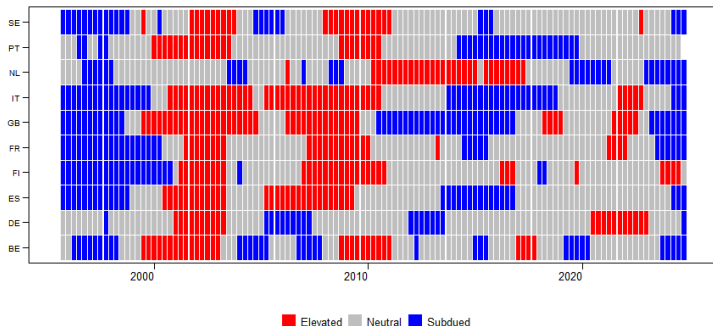
PCA (solid), QML (dashed), TSTEP (dotted) estimates. Shaded:  
ESRB-identified systemic banking crisis periods.

## Backup: FC — Early Warning AUROC (Full Table)

| Country | AUROC |      |       |      | Usefulness |      |       |      |
|---------|-------|------|-------|------|------------|------|-------|------|
|         | PCA   | QML  | TSTEP | BG   | PCA        | QML  | TSTEP | BG   |
| BE      | 0.54  | 0.56 | 0.55  | 0.58 | 0.20       | 0.20 | 0.21  | 0.29 |
| DE      | 0.65  | 0.50 | 0.43  | 0.46 | 0.17       | 0.14 | 0.06  | 0.18 |
| ES      | 0.94  | 0.85 | 0.91  | 0.51 | 0.45       | 0.40 | 0.43  | 0.23 |
| FR      | 0.45  | 0.63 | 0.57  | 0.62 | 0.10       | 0.23 | 0.17  | 0.31 |
| GB      | 0.61  | 0.52 | 0.59  | 0.64 | 0.27       | 0.23 | 0.28  | 0.32 |
| IT      | 0.90  | 0.98 | 0.92  | 0.53 | 0.37       | 0.47 | 0.37  | 0.26 |
| NL      | 0.68  | 0.49 | 0.61  | 0.57 | 0.25       | 0.15 | 0.20  | 0.29 |
| PT      | 0.52  | 0.60 | 0.56  | 0.51 | 0.23       | 0.25 | 0.26  | 0.26 |
| SE      | 0.65  | 0.53 | 0.67  | 0.65 | 0.17       | 0.13 | 0.18  | 0.32 |
| Avg     | 0.66  | 0.63 | 0.65  | 0.56 | 0.25       | 0.24 | 0.24  | 0.27 |

FI excluded (no systemic banking crisis in sample). BG = Basel gap.

## Backup: FC — Regimes Across Countries



Elevated (red), Neutral (grey), Subdued (blue) periods by country.